# Force-Receptive Scattering Transformation for Bilateral Teleoperations

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**Abstract**

A new force-receptive scattering transformation is proposed in this paper to enable independent contact force feedback for a bilateral teleoperation system, thereby enhancing teleoperation transparency in the presence of constant communication delays. The key idea is to increase the total number of data ports and scattering weights compared with the classical scattering transformation. These redundant weights are then designed and optimized to satisfy both passivity and steady-state transmission requirements. Non-passive human and environmental contact forces are also taken into account. The stability analysis is conducted entirely within the passivity framework, whose result is independent of the magnitude of the time delay. Under mild assumptions on the contact forces, both position and force tracking errors between the local and remote robots asymptotically converge to zero.

*Keywords:* telerobotics, teleoperation, scattering transformation, time delay.

## 1. Introduction

### 1.1. Background

In 1885, Heaviside published a paper in which he formulated the equation of electrical transmission lines (Donaghy-Spargo, 2018). His initial purpose was to explain complex phenomena of time delay and wave reflection occurring in long electrical lines (e.g., telegraph, telephone, power lines). Over one hundred years later, Anderson and Spong discovered that the lossless transmission line model of Heaviside can be used to transmit mechanical power in a passive manner over the Internet subject to communication time delays, via so-called scattering transformation (Anderson and Spong, 1989). This major breakthrough opened the gate for the boom in telerobotics researches from early 2000s onward. An array of new techniques has been invented since then to improve both stability and transparency for teleoperation systems: four-channel approach (Lawrence, 1993), wave-based control (Niemeyer and Slotine, 2004), time-domain passivity approach (TDPA) (Hannaford and Ryu, 2002), passivity-based control (Atashzar et al., 2017), passive-set-position-modulation (PSPM) framework (Lee and Huang, 2010), model-mediation control (Mitra and Niemeyer, 2008), energy-bounding algorithm (EBA) (Kim and Ryu, 2010), two-layer approach (Franken et al., 2011) (see survey papers of Hokayem and Spong, 2006; Nuño et al., 2011; Sun et al., 2014; Kebria et al., 2019 and the references therein for more details). Besides, the classical scattering transformation remains crucial in the toolbox when dealing with time delays. Therefore, a series of works was also proposed to improve the classical scattering transformation (Chopra et al., 2006; Miyoshi et al., 2006; Kanno and Yokokohji, 2012; Alise et al., 2009; Ferraguti et al., 2021). Notably, the notion of wave variable was introduced in Niemeyer and Slotine, 1991, by which the scattering transformation can be employed to passivate any two-port system whose $H_\infty$ norm is less than one. Early researches utilizing the scattering transformation in teleoperations often followed the modular design of passivity formalism via passivity theorem. A more recent approach was to use the scattering transformation under the framework of small-gain theorem. This trend was ignited by the work of Hirche et al., 2009 with the introduction of a generalized scattering transformation applied on network control systems and later extended to planar conic systems (Polushin, 2014) and non-planar conic ones (Usova et al., 2018). Recently, an upper strictly passive scattering transformation was established to enable the use of direct contact force feedback (instead of coordinating force feedback) to enhance transparency of teleoperations (Tho and Miyoshi, 2023).

### 1.2. Main contribution

To achieve better haptic transparency, a teleoperation system should be able to communicate two types of information between the local and remote sides: kinematics (positions and velocities of robots) and kinetics (actual contact forces). In the current literature, it is unclear how

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a scattering transformation can accommodate additional kinetic information along with kinematic data in a way that ensures passivity for both the delayed communication channel and the closed-loop teleoperation system. If a classical scattering transformation is employed, in order to transmit both kinematic and kinetic data over the network, one needs to mix them in the form of a weighted sum before submitting to the scattering transformation blocks as inputs (Aziminejad et al., 2008). Besides its unintuitive nature, such a technique poses a significant difficulty in proving or ensuring stability for the closed-loop system. In order to overcome such a problem, this paper puts forward a new force-receptive scattering transformation, which is capable of receiving both kinematic and kinetic information *independently* as inputs and modulating the outputs appropriately so that the communication channel subject to constant time delays become passive, while the integrity of data transmission in the steady state is maintained. This is the main contribution of the paper. Moreover, the majority of existing studies assumed that both human operator and environment are passive, which is not always true in practice. This paper will analyze the stability and transparency of a teleoperation system that uses the force-receptive scattering transformation when both human and environmental forces have non-passive (unknown) constant as well as time-varying components.

*Notations:* $\mathcal{L}_\infty$ and $\mathcal{L}_2$ denote the space of uniformly bounded functions and square integrable functions, respectively (Khalil, 2002, p. 196). The set of real numbers, non-negative real numbers, and non-negative integers are represented by $\mathbb{R}$, $\mathbb{R}^+$ and $\mathbb{N}$, respectively. The $n$-dimensional Euclidean space is $\mathbb{R}^n$. The zero matrix is denoted as $\mathbf{0}$ with an appropriate dimension. Moreover, let $\mathcal{N}$ denote the set $\{1, 2, 3\}$. The argument of any time-dependent function is the time variable $t$, except where it is delayed or explicitly written otherwise.

## 2. Force-receptive scattering transformation: design and analysis

We firstly introduce the structure of the proposed force-receptive scattering transformation and then devise appropriate constraints for scattering weights. The solvability of the force-receptive scattering transformation is then discussed. Afterward, the force-receptive scattering transformation is implemented in a teleoperation system with environmental contact force feedback to enhance the haptic transparency. Bounded-input-bounded-output (BIBO) and asymptotic stability of the force-receptive scattering transformation are later analyzed.

To clearly explain the basic concept, this section is devoted to designing a force-receptive scattering transformation for a three-channel teleoperation system. The proposed scattering transformation framework for other teleoperation architectures with four or more channels can be

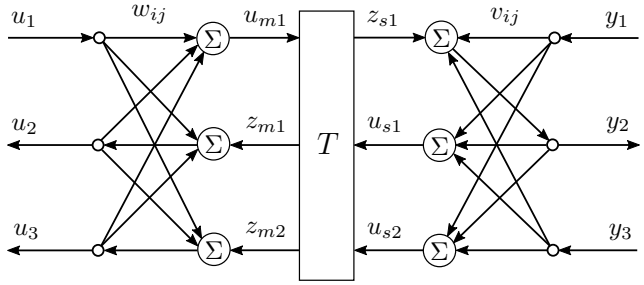


Figure 1: The force-receptive scattering transformation.

adapted in the same manner. The force-receptive scattering transformation structure is shown in Fig. 1, where the total number of ports and weights are increased to 6 and 18, respectively. The to-be-determined variables are the weights $w_{ij}$ and $v_{ij}$ $(i, j \in \mathcal{N})$. The local (or master) side sends a signal $u_1$ and receives the signals $u_2$ and $u_3$. The remote (or slave) side transmits the signals $y_1$ and $y_3$, and it also acquires a signal $y_2$. All sending and receiving signals are subjected to a constant time delay $T$, namely

$$\begin{aligned} z_{s1}(t) &= u_{m1}(t-T) \\ z_{m1}(t) &= u_{s1}(t-T) \\ z_{m2}(t) &= u_{s2}(t-T). \end{aligned} \tag{1}$$

The force-receptive scattering transformation needs to fulfill two requirements: passivity and steady-state transmission condition (see Definition 1). These requirements can be realized by devising proper scattering weights $w_{ij}$ and $v_{ij}$, as elaborated in the next subsections.

### 2.1. Passivity enforcement

To derive suitable constraints for the weights $w_{ij}$ and $v_{ij}$ so that the communication channel (orchestrated by the force-receptive scattering transformation) is passive, the following Lyapunov-Krasovskii storage function is chosen

$$V_c = \int_{t-T}^{t} \left[u_{m1}^2(\sigma) + u_{s1}^2(\sigma) + u_{s2}^2(\sigma)\right] d\sigma. \tag{2}$$

By using (1), the first derivative of $V_c$ can be obtained as

$$\dot{V}_c = u_{m1}^2(t) + u_{s1}^2(t) + u_{s2}^2(t) - z_{m1}^2(t) - z_{m2}^2(t) - z_{s1}^2(t). \tag{3}$$

From Fig. 1, the following relations can be inferred

$$\begin{aligned} u_{m1} &= w_{11}u_1 + w_{21}u_2 + w_{31}u_3 \\ z_{m1} &= w_{12}u_1 + w_{22}u_2 + w_{32}u_3 \\ z_{m2} &= w_{13}u_1 + w_{23}u_2 + w_{33}u_3 \\ z_{s1} &= v_{11}y_1 + v_{21}y_2 + v_{31}y_3 \\ u_{s1} &= v_{12}y_1 + v_{22}y_2 + v_{32}y_3 \\ u_{s2} &= v_{13}y_1 + v_{23}y_2 + v_{33}y_3. \end{aligned} \tag{4}$$

Substituting (4) into (3) and rearranging the result yield

$$\dot{V}_c = \left(w_{11}^2 - \sum_{k=2}^{3} w_{1k}^2\right) u_1^2 + \left(w_{21}^2 - \sum_{k=2}^{3} w_{2k}^2\right) u_2^2$$

$$
\begin{aligned}
&+\left(w_{31}^2-\sum_{k=2}^{3} w_{3k}^2\right) u_3^2+\left(\sum_{k=2}^{3} v_{1k}^2-v_{11}^2\right) y_1^2 \\
&+\left(\sum_{k=2}^{3} v_{2k}^2-v_{21}^2\right) y_2^2+\left(\sum_{k=2}^{3} v_{3k}^2-v_{31}^2\right) y_3^2 \\
&+2\left(w_{11} w_{21}-\sum_{k=2}^{3} w_{1k} w_{2k}\right) u_1 u_2 \\
&+2\left(w_{11} w_{31}-\sum_{k=2}^{3} w_{1k} w_{3k}\right) u_1 u_3 \\
&+2\left(w_{21} w_{31}-\sum_{k=2}^{3} w_{2k} w_{3k}\right) u_2 u_3 \\
&+2\left(\sum_{k=2}^{3} v_{1k} v_{2k}-v_{11} v_{21}\right) y_1 y_2 \\
&+2\left(\sum_{k=2}^{3} v_{1k} v_{3k}-v_{11} v_{31}\right) y_1 y_3 \\
&+2\left(\sum_{k=2}^{3} v_{2k} v_{3k}-v_{21} v_{31}\right) y_2 y_3 .
\end{aligned} \tag{5}
$$

Based on (5), the force-receptive scattering transformation passivates the communication channel with the passive input–output pair $\boldsymbol{u}$–$\boldsymbol{y}$ if the following constraints are fulfilled

$$
\begin{aligned}
&w_{11}^2-\sum_{k=2}^{3} w_{1k}^2=\gamma_m,\ w_{21}^2-\sum_{k=2}^{3} w_{2k}^2=0,\ w_{31}^2-\sum_{k=2}^{3} w_{3k}^2=0 \\
&\sum_{k=2}^{3} v_{1k}^2-v_{11}^2=\gamma_s,\ \sum_{k=2}^{3} v_{2k}^2-v_{21}^2=0,\ \sum_{k=2}^{3} v_{3k}^2-v_{31}^2=0, \\
&w_{11} w_{21}-\sum_{k=2}^{3} w_{1k} w_{2k}=\delta_1,\ w_{11} w_{31}-\sum_{k=2}^{3} w_{1k} w_{3k}=\delta_2, \\
&w_{21} w_{31}-\sum_{k=2}^{3} w_{2k} w_{3k}=\delta_3,\ \sum_{k=2}^{3} v_{1k} v_{2k}-v_{11} v_{21}=\Delta_1, \\
&\sum_{k=2}^{3} v_{1k} v_{3k}-v_{11} v_{31}=\Delta_2,\ \sum_{k=2}^{3} v_{2k} v_{3k}-v_{21} v_{31}=\Delta_3,
\end{aligned} \tag{6}
$$

where $\boldsymbol{u} := [\delta_1 u_1 \ \ \delta_2 u_1 \ \ \delta_3 u_2 \ \ \Delta_1 y_1 \ \ \Delta_2 y_1 \ \ \Delta_3 y_2]^\top \in \mathbb{R}^6$ and $\boldsymbol{y} := 2\,[u_2 \ \ u_3 \ \ u_3 \ \ y_2 \ \ y_3 \ \ y_3]^\top \in \mathbb{R}^6$. Note that, $\delta_i$ and $\Delta_i$ $(i \in \mathcal{N})$ are also to-be-determined parameters, which will be specified in section 2.4. In addition, $\gamma_m \leq 0$ and $\gamma_s \leq 0$ are the desired levels of excessive passivity at the ports $u_1$ and $y_1$, respectively. When the scattering weights $w_{ij}$ and $v_{ij}$ verify (6), it is true that

$$
\dot{V}_c = \boldsymbol{u}^\top \boldsymbol{y} + \gamma_m u_1^2 + \gamma_s y_1^2. \tag{7}
$$

Moreover, based on Fig. 1 and the relation (4), it is also required that $w_{22} \neq 0$, $w_{33} \neq 0$, and $v_{21} \neq 0$.

From Fig. 1, it is worth emphasizing that the inputs–outputs of the force-receptive scattering transformation for the local and remote sides are $(u_1, z_{m1}, z_{m2})$–$(u_{m1}, u_2, u_3)$ and $(z_{s1}, y_1, y_3)$–$(u_{s1}, u_{s2}, y_2)$, respectively.

### 2.2. Steady-state transmission condition enforcement

It is well known that to ensure passivity for the communication channel, the choice of the scattering weights $w_{ij}$ and $v_{ij}$ should not allow direct data transmission, i.e., the condition $u_2(t) = y_2(t-T)$ and $u_3(t) = y_3(t-T)$ $\forall t \geq 0$ are prohibited. Nevertheless, one should preserve certain transmission criteria in the steady state to guarantee the correctness of data transfer when everything settles down, so that the teleoperation system is well-behaved. For this reason, the following steady-state (SS) transmission requirement is defined.

**Definition 1 (SS transmission requirement).** Considering the communication channel with the force-receptive scattering transformation as shown in Fig. 1. Assuming that all input/output signals of the system are in their steady states, i.e., the following limits

$$
\lim_{t\to\infty} u_i(t) := \bar{u}_i,\ \lim_{t\to\infty} y_i(t) := \bar{y}_i \ \ (i \in \mathcal{N})
$$

exist and are finite, and the condition $\bar{u}_1 = \bar{y}_1 = 0$ holds. Then, the communication channel is said to satisfy the steady-state transmission requirement if and only if

$$
\bar{u}_i = \bar{y}_i \ \ (i = 2, 3)\,.
$$

The following Theorem summarizes the constraints on the weights $w_{ij}$ and $v_{ij}$ to realize the above-mentioned steady-state transmission requirement.

**Theorem 1.** *A necessary and sufficient condition for the force-receptive scattering transformation shown in Fig. 1 to fulfill the steady-state transmission requirement is*

$$
\begin{aligned}
&w_{2i} = v_{2i},\ w_{3i} = v_{3i}\ (i \in \mathcal{N}) \\
&\operatorname{rank}(\boldsymbol{A}) = 2,\ \textit{where}\ \boldsymbol{A} = \begin{bmatrix} w_{21} & w_{22} & w_{23} \\ w_{31} & w_{32} & w_{33} \end{bmatrix}^\top .
\end{aligned} \tag{8}
$$

Proof. The proof is along the same lines as Tho and Miyoshi (2023, Theorem 2).

### 2.3. Existence of solution

In this subsection, the solvability of the force-receptive scattering transformation will be discussed. To sum up the results of subsections 2.1 and 2.2, the weights $w_{ij}$ and $v_{ij}$ must fulfill the following set of polynomial constraints

$$
\begin{cases}
\bullet\ \ w_{22} \neq 0,\ w_{33} \neq 0,\ \text{and}\ v_{21} \neq 0 \\
\bullet\ \ \text{polynomial equations (6) (for passivity)} \\
\bullet\ \ \text{polynomial constr. (8) (for SS transmission).}
\end{cases} \tag{9}
$$

Excluding inequalities, the above set of polynomial constraints has 18 polynomial equations in a total of 26 free variables including $w_{ij}, v_{ij}, \delta_i, \Delta_i, \gamma_m$, and $\gamma_s$ $(i, j \in \mathcal{N})$. Here, we are only interested in real solutions of (9). Since the number of equations and variables in (9) is relatively large, the computation of Gröbner basis—the main tool

from algebraic geometry to study solutions of polynomial equations (Cox et al., 2005, p. 13)—is inefficient. Furthermore, it is next to impossible to obtain closed-form expressions for all solutions. However, we argue that (9) has at least one real solution.

**Theorem 2.** *Consider the system of polynomial constraints (9). If it is chosen that*

$$\Delta_3 = -\delta_3 \neq 0,\ \frac{2\delta_1\delta_2}{\delta_3} - \gamma_m \geq 0,\ \frac{2\Delta_1\Delta_2}{\delta_3} + \gamma_s \geq 0, \quad (10)$$

*then the force-receptive scattering transformation is solvable, i.e., (9) has at least one real solution of $w_{ij}$ and $v_{ij}$ $(i, j \in \mathcal{N})$.*

Proof. The proof is done by pointing out one instance of $w_{ij}$ and $v_{ij}$ that satisfies (9). Such an instance should be carefully selected to reduce the complexity in solving. After some trial and error, the following special solution of (9) is chosen

$$w_{22} = w_{21},\ w_{33} = -w_{31}.$$

Substituting this relation as well as the equations of (8) into (6) results in $\Delta_3 = -\delta_3$. Moreover, (9) can also be completely reduced to

$$\begin{aligned} &w_{11}^2 - w_{12}^2 - w_{13}^2 = \gamma_m,\ w_{21}\left(w_{11} - w_{12}\right) = \delta_1, \\ &w_{31}\left(w_{11} + w_{13}\right) = \delta_2,\ w_{21}w_{31} = \delta_3,\ v_{12}^2 + v_{13}^2 - v_{11}^2 = \gamma_s, \\ &w_{21}\left(v_{12} - v_{11}\right) = \Delta_1,\ -w_{31}\left(v_{13} + v_{11}\right) = \Delta_2, \\ &w_{23} = 0,\ w_{32} = 0,\ v_{2i} = w_{2i},\ v_{3i} = w_{3i}\ (i \in \mathcal{N}), \end{aligned} \quad (11)$$

in addition to the inequalities $w_{21} \neq 0$ and $w_{31} \neq 0$. Choosing $w_{21} := c \in \mathbb{R}\backslash\{0\}$, then from (11), we have

$$\begin{aligned} &w_{12} = w_{11} - \delta_1 c^{-1},\ w_{13} = -w_{11} + \delta_2\delta_3^{-1}c,\ w_{31} = \delta_3 c^{-1} \\ &v_{12} = v_{11} + \Delta_1 c^{-1},\ v_{13} = -v_{11} - \Delta_2\delta_3^{-1}c, \end{aligned}$$

where $w_{11}$ and $v_{11}$ are determined by

$$\begin{aligned} &w_{11}^2 - 2\left(\frac{\delta_1}{c} + \frac{\delta_2 c}{\delta_3}\right)w_{11} + \left(\frac{\delta_1}{c}\right)^2 + \left(\frac{\delta_2 c}{\delta_3}\right)^2 + \gamma_m = 0, \\ &v_{11}^2 + 2\left(\frac{\Delta_1}{c} + \frac{\Delta_2 c}{\delta_3}\right)v_{11} + \left(\frac{\Delta_1}{c}\right)^2 + \left(\frac{\Delta_2 c}{\delta_3}\right)^2 - \gamma_s = 0. \end{aligned}$$

It is clear that, in order to have real solutions of $w_{11}$ and $v_{11}$, the discriminants of the above quadratic equations need to be non-negative, namely that

$$\frac{2\delta_1\delta_2}{\delta_3} - \gamma_m \geq 0,\ \frac{2\Delta_1\Delta_2}{\delta_3} + \gamma_s \geq 0.$$

Moreover, it is required that $w_{31} \neq 0$, thus we need $\delta_3 \neq 0$ since $w_{31} = \delta_3/c$. This therefore completes the proof.

In summary, the choice for $\delta_i$, $\Delta_i$, $\gamma_m$, and $\gamma_s$ $(i \in \mathcal{N})$ should strictly adhere to (10) to guarantee the solvability of the force-receptive scattering transformation. Note also that (9) has more free variables than constraints; thus, it can reasonably be expected that multiple solutions of the scattering weights $w_{ij}$ and $v_{ij}$ might exist. Some solutions might result in better control performance for the force-receptive scattering transformation than others. Therefore, in practice, numerical solvers can be employed to fully explore all possibilities. The aforementioned parameter redundancy is a feature of the force-receptive scattering transformation that can be exploited by some optimization/training procedures to improve teleoperation transparency (see section 4).

### *2.4. Teleoperation using the force-receptive scattering transformation*

In this subsection, we introduce a three-channel teleoperation system employing the force-receptive scattering transformation with environmental contact force feedback as shown in Fig. 2. Note that, the contact force (i.e., $F_e$) is transmitted independently from the kinematic information. In order to keep the subsequent analyses as simple as possible, one degree-of-freedom local and remote robots will be considered as similar to Anderson and Spong, 1989 and Tho and Miyoshi, 2023. The dynamics of the local robot is given by

$$\ddot{x}_m = \tau_m + F_h, \quad (12)$$

where $x_m$, $\tau_m$, and $F_h$ denote the position, the applied motor force, and the applied human force of the local robot, respectively. Similarly, the remote robot is specified by

$$\ddot{x}_s = \tau_s - F_e, \quad (13)$$

where $x_s$, $\tau_s$, and $F_e$ represent the position, the applied motor force, and the environmental contact force of the remote robot.

Let $r_m := \dot{x}_m + \lambda x_m$ and $r_s := \dot{x}_s + \lambda x_s$ $(\lambda > 0)$[1]. The local and remote robots are, respectively, controlled by[2]

$$\tau_m = -\lambda\dot{x}_m + K_m\left(r_{md} - r_m\right) - F_{md} \quad (14)$$
$$\tau_s = -\lambda\dot{x}_s + K_s\left(r_{sd} - r_s\right) + \dot{r}_{sd} + F_e, \quad (15)$$

where $K_m, K_s > 0$ and local force feedback has been used on $\tau_s$ (see Hashtrudi-Zaad and Salcudean, 2002 and Naerum and Hannaford, 2009). Note that, the signals $r_{md}$, $r_{sd}$, and $F_{md}$ are the outputs of the force-receptive scattering transformation when the following signal substitutions are performed on the general force-receptive scattering transformation shown in Fig. 1

$$\begin{aligned} &u_1 = r_m - r_{md},\ u_2 = r_{md},\ u_3 = F_{md}, \\ &y_1 = r_s - r_{sd},\ y_2 = r_{sd},\ y_3 = F_e. \end{aligned} \quad (16)$$

[1] The composite signals $r_m$ and $r_s$ were first introduced in Chopra and Spong, 2004.

[2] Since $\tau_s$ requires local force feedback, a force sensor is needed to measure the actual environmental contact force $F_e$.

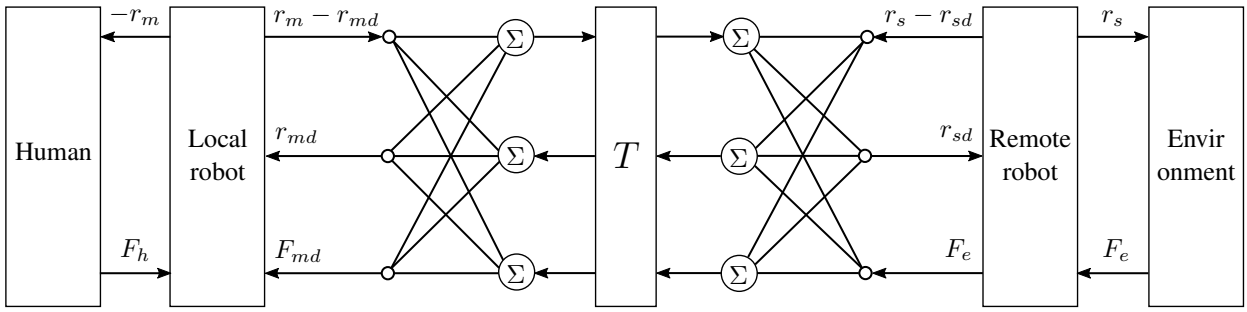


Figure 2: A three-channel teleoperation architecture using the force-receptive scattering transformation.

By substituting (14)–(15) into (12)–(13), the closed-loop systems of the local and remote sides are, respectively, given by

$$\dot{r}_m = -K_m (r_m - r_{md}) - F_{md} + F_h \tag{17}$$
$$\dot{r}_s - \dot{r}_{sd} = -K_s (r_s - r_{sd}) . \tag{18}$$

From (18), the results $r_s - r_{sd} \to 0$ and $\dot{r}_s - \dot{r}_{sd} \to 0$ as $t \to \infty$ can immediately be obtained. Moreover, $r_s - r_{sd} \in \mathcal{L}_\infty$ and $\dot{r}_s - \dot{r}_{sd} \in \mathcal{L}_\infty$.

The stability of the overall teleoperation system including the local side, the communication channel operated by the force-receptive scattering transformation, and the remote side can be analyzed by using the Lyapunov-Krasovskii functional $V = 0.5 r_m^2 + 0.5 (r_s - r_{sd})^2 + 0.5 V_c$, where $V_c$ is given by (2), that is to say

$$V = \frac{1}{2} r_m^2 + \frac{1}{2} (r_s - r_{sd})^2 + \frac{1}{2} \int_{t-T}^{t} \left( u_{m1}^2 + u_{s1}^2 + u_{s2}^2 \right) d\sigma. \tag{19}$$

By using (7) and (17)–(18), the first derivative of $V$ is

$$\begin{aligned} \dot{V} = & - K_m r_m (r_m - r_{md}) - F_{md} r_m + F_h r_m - K_s (r_s - r_{sd})^2 \\ & + \delta_1 (r_m - r_{md}) r_{md} + \delta_2 (r_m - r_{md}) F_{md} \\ & + \delta_3 F_{md} r_{md} + \Delta_1 (r_s - r_{sd}) r_{sd} + \Delta_2 (r_s - r_{sd}) F_e \\ & - \delta_3 F_e r_{sd} + \gamma_m (r_m - r_{md})^2 + \gamma_s (r_s - r_{sd})^2 . \end{aligned}$$

Rearranging the previous equation yields

$$\begin{aligned} \dot{V} = & \ (r_m - r_{md}) (\delta_1 r_{md} - K_m r_m) \\ & + [(\delta_2 - 1) r_m - (\delta_2 - \delta_3) r_{md}] F_{md} + F_h r_m \\ & + \Delta_1 (r_s - r_{sd}) r_{sd} - (\Delta_2 + \delta_3) F_e r_{sd} + \Delta_2 F_e r_s \\ & - K_s (r_s - r_{sd})^2 + \gamma_m (r_m - r_{md})^2 + \gamma_s (r_s - r_{sd})^2 . \end{aligned} \tag{20}$$

Based on (20), one can choose

$$\begin{aligned} & \delta_1 = K_m, \ \delta_2 = 1, \delta_3 = \delta_2 = 1, \\ & \Delta_1 = 0, \ \Delta_2 = \Delta_3 = -\delta_3 = -1, \ \gamma_m = \gamma_s = 0, \end{aligned} \tag{21}$$

to obtain the following result

$$\dot{V} = -K_m (r_m - r_{md})^2 - K_s (r_s - r_{sd})^2 + F_h r_m - F_e r_s. \tag{22}$$

**Remark 1.** It can straightforwardly be checked that (21) satisfies the condition (10), and thus, the solvability of the force-receptive scattering transformation is always guaranteed while the validity of (22) is maintained.

If $(F_h, -r_m)$ and $(F_e, r_s)$ define passive maps, namely that

$$\int_0^t (-F_h r_m + F_e r_s) d\sigma \geq 0,$$

then from (22), the following result holds

$$\begin{aligned} & \frac{d}{dt} \left[ V + \int_0^t (-F_h r_m + F_e r_s) d\sigma \right] \\ & \quad = -K_m (r_m - r_{md})^2 - K_s (r_s - r_{sd})^2 \leq 0. \end{aligned} \tag{23}$$

Therefore, the stability of the closed-loop teleoperation system can easily be concluded. However, the assumption that both human and remote environment are passive does not generally hold in practice since the human operator surely generates energy to operate the whole system. More specifically, the human force $F_h$ consists of both feedback and feedforward components that behave similarly to a spring-damper system driven by a time-dependent exogenous input (Yang et al., 2011). To account for a dynamic remote environment, a similar impedance model should be utilized for the environmental force $F_e$. For this reason, throughout the rest of the paper, we will analyze the system stability for the case that both human and environmental contact forces contain non-passive constant as well as time-varying components, that is to say, $F_h$ and $F_e$ are given by the following expressions

$$\begin{aligned} F_h(t) &= -K_h [r_m(t) - d_m] + f_h^\star(t) \\ F_e(t) &= K_e [r_s(t) - d_s] + f_e^\star(t), \end{aligned} \tag{24}$$

where $K_h > 0$ and $K_e \geq 0$ are the spring coefficients of the human hand and the environment, respectively. Moreover, the constant forces are $d_m, d_s \in \mathbb{R}$, whereas the time-varying forces $f_h^\star(t)$ and $f_e^\star(t)$ are supposed to be differentiable almost everywhere, and their derivatives are continuous almost everywhere. It is also assumed that $f_h^\star, f_e^\star \in \mathcal{L}_\infty$ and $\dot{f}_h^\star, \dot{f}_e^\star \in \mathcal{L}_2 \cap \mathcal{L}_\infty$. From (24), the feedback and feedforward parts of the human force $F_h$ are $-K_h [r_m(t) - d_m]$ and $f_h^\star(t)$, respectively. The term $d_m$

is a constant reference of the teleoperation task. A similar interpretation can be applied for the environmental force $F_e$. Note that, (24) is indeed a generalization of the force models employed in Chopra et al., 2008 and Lozano et al., 2007.

We end this subsection by pointing out that if the variable selection (21) is kept, then $v_{1i} = w_{2i}$ $(i \in \mathcal{N})$ follows. This result is essential for simplifying the overall expression of the force-receptive scattering communication law in the next subsection.

**Proposition 1.** *The choice of parameters $\delta_i, \Delta_i, \gamma_m, \gamma_s$ $(i \in \mathcal{N})$ made in (21) results in the relation $v_{1i} = w_{2i}$ $(i \in \mathcal{N})$ for the solution of polynomial constraints (9).*

Proof. The result can be proved by performing the following trigonometric substitutions

$$w_{i2} = a_i \sin\theta_i,\ w_{i3} = a_i \cos\theta_i,\ (i \in \mathcal{N})$$
$$v_{12} = a_4 \sin\theta_4,\ v_{13} = a_4 \cos\theta_4,$$

where $a_i, \theta_i \in \mathbb{R}$ $(i = 1, \ldots, 4)$. Applying these substitutions as well as (21) on (9) yields

$$a_i a_3 \left[1 - \cos\left(\theta_i - \theta_3\right)\right] = 1,\ (i = 1, 2, 4) \tag{25}$$
$$a_2 a_4 \left[1 - \cos\left(\theta_2 - \theta_4\right)\right] = 0. \tag{26}$$

It can be inferred from (26) that $a_2 = 0$, or $a_4 = 0$, or $\theta_2 = \theta_4 + 2k\pi$ $(k \in \mathbb{N})$. The two cases $a_2 = 0$ and $a_4 = 0$ can immediately be rejected. Substituting the remaining case $\theta_2 = \theta_4 + 2k\pi$ into (25) leads to

$$a_2 a_3 \left[1 - \cos\left(\theta_2 - \theta_3\right)\right] = 1$$
$$a_3 a_4 \left[1 - \cos\left(\theta_2 - \theta_3\right)\right] = 1.$$

As a consequence, $a_2 = a_4$. Therefore, the results $v_{12} = w_{22}$, $v_{13} = w_{23}$, and $v_{11} = w_{21}$ can easily be concluded. This completes the proof.

### 2.5. Stability of the force-receptive scattering communication law

Substituting the relation (4) into (1) and employing the result $v_{1i} = w_{2i}$ $(i \in \mathcal{N})$ of Proposition 1 yield

$$\begin{aligned} u_2(t) - y_2(t-T) &= y_1(t-T) + \omega_2 u_1(t), \\ y_2(t-2T) - y_2(t) &= \left[y_1(t) - y_1(t-2T)\right] \\ &\quad + w_{21}^{-1}(\omega_3 - w_{21}\omega_2) u_1(t-T) \\ &\quad + w_{21}^{-1} w_{31} \left[y_3(t) - y_3(t-2T)\right], \\ u_3(t) - y_3(t-T) &= \omega_1 \left(w_{12}w_{23} - w_{13}w_{22}\right) u_1(t), \end{aligned} \tag{27}$$

where it is denoted that $\omega_1 := (w_{22}w_{33} - w_{23}w_{32})^{-1}$, $\omega_2 := \omega_1(w_{13}w_{32} - w_{12}w_{33})$, and $\omega_3 := -w_{11} + w_{31}\omega_1(w_{13}w_{22} - w_{12}w_{23})$. By performing the signal substitution (16) on (27), it is true that

$$r_{md}(t) - r_m(t) = -(1+\omega_2)^{-1}\left[r_m(t) - r_s(t-T)\right], \tag{28}$$
$$F_{md}(t) - F_e(t-T) = \omega_1 \left(w_{12}w_{23} - w_{13}w_{22}\right)\left[r_m(t) - r_{md}(t)\right], \tag{29}$$

$$\begin{aligned} r_s(t) + \omega_4 r_m(t-T) &= (1+\omega_4) r_s(t-2T) \\ &\quad - w_{21}^{-1} w_{31}\left[F_e(t) - F_e(t-2T)\right], \end{aligned} \tag{30}$$

in which $\omega_4 := \left(w_{21}\left(1+\omega_2\right)\right)^{-1}\left(\omega_3 - w_{21}\omega_2\right)$. If the environmental force $F_e$ is given by (24), i.e., $F_e = K_e(r_s - d_s) + f_e^\star$, then (30) becomes

$$\begin{aligned} r_s(t) &= \varepsilon r_s(t-2T) + (1-\varepsilon) r_m(t-T) \\ &\quad + \omega_5\left[f_e^\star(t) - f_e^\star(t-2T)\right], \end{aligned} \tag{31}$$

where $\omega_5 := -\left(w_{21} + K_e w_{31}\right)^{-1} w_{31}$ and

$$\varepsilon = \frac{1 + w_{21}^{-1} w_{31} K_e + \omega_4}{1 + w_{21}^{-1} w_{31} K_e} \quad (\text{with } K_e \geq 0). \tag{32}$$

To analyze the stability of the force-receptive scattering communication law, one needs to study the stability of the algebraic delayed equation (31). We firstly recall the following lemma.

**Lemma 1** (Theorem 3, Tho and Miyoshi, 2023)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function with respect to its argument $t$. Suppose that $f$ is bounded for all $t < 0$. If $f$ satisfies the following algebraic delayed equation (with $T \geq 0$)*

$$f(t) = \beta f(t-2T) + \xi(t),$$

*where $|\beta| < 1$ $(\beta \in \mathbb{R})$ and $\xi(t) \to 0$ as $t \to \infty$, then $f(t) \in \mathcal{L}_\infty$ and $f(t) \to 0$ as $t \to \infty$.*

The BIBO and asymptotic stability of the force-receptive scattering communication law are summarized as follows.

**Theorem 3.** *Consider the algebraic delayed equation (31). Assume that $r_s(t)$ is bounded for all $t < 0$ and that $|\varepsilon| < 1$ for all $K_e \geq 0$. The following results hold*

1. *(BIBO stability) If $r_m \in \mathcal{L}_\infty$ and $r_s - r_{sd} \in \mathcal{L}_\infty$, then $r_s \in \mathcal{L}_\infty$ and the outputs of the force-receptive scattering transformation including $r_{md}$, $r_{sd}$, and $F_{md}$ also belong to $\mathcal{L}_\infty$.*

2. *(Asymptotic stability) If $r_m(t) - r_m(t-2T) \to 0$ and $f_e^\star(t) - f_e^\star(t-2T) \to 0$ as $t \to \infty$, then $r_m(t-T) - r_s(t) \to 0$, $r_{md}(t) - r_m(t) \to 0$, and $F_{md}(t) - F_e(t-T) \to 0$ as $t \to \infty$.*

Proof. We firstly prove 1). By assumption, we have $r_m, f_e^\star \in \mathcal{L}_\infty$ and $|\varepsilon| < 1$. Therefore, the application of Lemma 1 on (31) immediately yields $r_s \in \mathcal{L}_\infty$, and the result $r_{md} \in \mathcal{L}_\infty$ follows by using (28). Since $F_e = K_e(r_s - d_s) + f_e^\star$ and $r_s, f_e^\star \in \mathcal{L}_\infty$, we have $F_e \in \mathcal{L}_\infty$. Consequently, $F_{md} \in \mathcal{L}_\infty$ by (29). From the assumption $r_s - r_{sd} \in \mathcal{L}_\infty$, it is true that $r_{sd} \in \mathcal{L}_\infty$ because $r_s \in \mathcal{L}_\infty$.

To prove the statement 2), we rewrite (31) as follows

$$\begin{aligned} r_m(t-T) - r_s(t) &= \varepsilon\left[r_m(t-3T) - r_s(t-2T)\right] \\ &\quad + \varepsilon\left[r_m(t-T) - r_m(t-3T)\right] \\ &\quad - \omega_5\left[f_e^\star(t) - f_e^\star(t-2T)\right]. \end{aligned}$$

Let $e_p(t) := r_m(t-T) - r_s(t)$, the above equation becomes

$$e_p(t) = \varepsilon e_p(t-2T) + \varepsilon\left[r_m(t-T) - r_m(t-3T)\right] - \omega_5\left[f_e^\star(t) - f_e^\star(t-2T)\right]. \quad (33)$$

By applying Lemma 1 on (33) and using the assumptions that $r_m(t) - r_m(t-2T) \to 0$ and $f_e^\star(t) - f_e^\star(t-2T) \to 0$ as $t \to \infty$, we have $e_p(t) \to 0$, and thus $r_m(t-T) - r_s(t) \to 0$ as $t \to \infty$. From (28)–(29), it can be obtained that $r_{md}(t) - r_m(t) \to 0$ and $F_{md}(t) - F_e(t-T) \to 0$ as $t \to \infty$. This therefore completes the proof.

**Remark 2.** Generally, the fulfillment of the requirement $|\varepsilon| < 1$ for all $K_e \geq 0$ depends on the values of the resulting weights $w_{ij}$ and $v_{ij}$ of the force-receptive scattering transformation. However, we can show that there is always one instance of $w_{ij}$ and $v_{ij}$ that makes $|\varepsilon| < 1$ for all $K_e \geq 0$. Continuing the proof of Theorem 2 on the solvability of the force-receptive scattering transformation, we have $\omega_3 = -w_{21}$ if the variable selection (21) is kept. Since $\omega_4 = (w_{21}(1+\omega_2))^{-1}(\omega_3 - w_{21}\omega_2)$, it is true that $\omega_4 = -1$. Therefore, $\varepsilon = K_e w_{21}^{-1} w_{31}\left(1 + w_{21}^{-1} w_{31} K_e\right)^{-1} = K_e c^{-2}\left(1 + K_e c^{-2}\right)^{-1}$, where the results $w_{21} = c \neq 0$ and $w_{31} = \delta_3/c = 1/c$ have been used. It is clear that, in this case, $|\varepsilon| < 1$ for all $K_e \geq 0$.

## 3. Stability and transparency analysis

In this section, the stability and transparency of the overall teleoperation system (including the local robot, the communication channel, and the remote robot) will be analyzed. Note that, the human force $F_h$ and the environmental force $F_e$ with non-passive components are given in (24). We firstly introduce two technical lemmas. The first lemma is essential for proving the boundedness of all signals in the closed-loop teleoperation system, whereas the second lemma will be employed to show the asymptotic convergence of the force and position tracking errors between the two robots.

### *3.1. Technical lemmas*

**Lemma 2.** *Let $K$, $d$, and $D$ be real numbers and $K > 0$. If a continuous function $f : \mathbb{R}^+ \to \mathbb{R}^+$ satisfies the following integral inequality*

$$-\int_0^t (Kf(\tau) - d)\, f(\tau) d\tau + D \geq 0 \quad \forall t \geq 0, \quad (34)$$

*then $f$ belongs to $\mathcal{L}_\infty$.*

Proof. Appendix A.

**Lemma 3.** *Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function that is differentiable almost everywhere on $\mathbb{R}$. Suppose that $\dot{f} \in \mathcal{L}_2 \cap \mathcal{L}_\infty$ and $\dot{f}$ is continuous almost everywhere, then*

$$f(t) - f(t-q) \to 0 \text{ as } t \to \infty \quad \forall q \geq 0.$$

Proof. Appendix B.

### *3.2. Stability analysis*

The local robot either operates in the contact or non-contact (free-motion) mode. In the contact mode, $K_e > 0$, whereas in the free-motion mode, $F_e(t) \equiv 0$ (i.e., $K_e = 0$ and $f_e^\star(t) \equiv 0$). Both the position and velocity of the local and remote robots are supposed to be uniformly bounded for all $t < 0$. The stability result of the closed-loop teleoperation system is summarized as follows.

**Theorem 4.** *Consider the teleoperation system shown in Fig. 2. Assume that the local and remote robots (12)–(13), subject to the applied human and environmental forces (24), are controlled by the control laws (14)–(15). Moreover, the communication channel is orchestrated by the force-receptive scattering communication law (28)–(30), whose weights $w_{ij}$ and $v_{ij}$ $(i, j \in \mathcal{N})$ are determined by (9) with the variable selection (21) so that $|\varepsilon| < 1$ for all $K_e \geq 0$ ($\varepsilon$ is defined in (32)). Then, all signals of the closed-loop teleoperation system are uniformly bounded.*

Proof. It was shown in section 2.4 that if the following Lyapunov-Krasovskii functional

$$V = \frac{1}{2} r_m^2 + \frac{1}{2}(r_s - r_{sd})^2 + \frac{1}{2}\int_{t-T}^t \left(u_{m1}^2 + u_{s1}^2 + u_{s2}^2\right) d\sigma$$

is chosen, then it is true that

$$\dot{V} = -K_m(r_m - r_{md})^2 - K_s(r_s - r_{sd})^2 + F_h r_m - F_e r_s.$$

Substituting the applied human and environmental forces (24) into the previous equation leads to the result that

$$\dot{V} \leq -K_h r_m^2 - K_e r_s^2 + \left(|f_h^\star| + |K_h d_m|\right)|r_m| + \left(|f_e^\star| + |K_e d_s|\right)|r_s|.$$

Since $f_h^\star, f_e^\star \in \mathcal{L}_\infty$, $K_h > 0$, $K_e \geq 0$, and $d_m, d_s \in \mathbb{R}$, there exists $D_m \geq 0$ and $D_s \geq 0$ such that $|f_h^\star| + |K_h d_m| \leq D_m$ and $|f_e^\star| + |K_e d_s| \leq D_s$. Therefore

$$\dot{V} \leq -K_h r_m^2 - K_e r_s^2 + D_m|r_m| + D_s|r_s|. \quad (35)$$

In the free-motion mode ($f_e^\star \equiv 0$ and $K_e = 0$), we have $D_s = 0$, and thus, (35) becomes $\dot{V} \leq -K_h r_m^2 + D_m|r_m|$. On the other hand, when the remote robot is in the contact mode ($K_e > 0$), it is true that

$$\dot{V} \leq -\min(K_h, K_e)\left(r_m^2 + r_s^2\right) + \sqrt{D_m^2 + D_s^2}\sqrt{r_m^2 + r_s^2}.$$

For this reason, in both free-motion and contact modes, the following result holds

$$\dot{V} \leq -\widetilde{K} r^2 + \widetilde{D} r, \quad (36)$$

where it is denoted that $r := |r_m|$, $\widetilde{K} := K_h$, and $\widetilde{D} := D_m$ in the free-motion mode, and that $r := \sqrt{r_m^2 + r_s^2}$, $\widetilde{K} := \min(K_h, K_e)$, $\widetilde{D} := \sqrt{D_m^2 + D_s^2}$ in the contact mode. Integrating (36) once yields

$$0 \leq V(t) \leq -\int_0^t (\widetilde{K} r(\tau) - \widetilde{D}) r(\tau) d\tau + V(0). \quad (37)$$

By applying Lemma 2 on (37), the result $r \in \mathcal{L}_\infty$ can be obtained, leading to the fact that $r_m \in \mathcal{L}_\infty$. Since $r_m \in \mathcal{L}_\infty$, $r_s - r_{sd} \in \mathcal{L}_\infty$ (see (18)), $|\varepsilon| < 1\ \forall K_e \geq 0$ (by assumption), and $r_s$ is bounded for all $t < 0$ (by assumption), it can be inferred that $r_s$, $r_{md}$, $r_{sd}$, and $F_{md}$ belong to $\mathcal{L}_\infty$, as a result of the statement 1) of Theorem 3. By using (24), the result $F_h, F_e \in \mathcal{L}_\infty$ follows, meaning that both applied human and environmental forces are uniformly bounded. Because $r_i = \dot{x}_i + \lambda x_i$ ($\lambda > 0$) and $r_i \in \mathcal{L}_\infty$ ($i = m, n$), we also have that $x_m$, $\dot{x}_m$, $x_s$, $\dot{x}_s \in \mathcal{L}_\infty$, i.e., the positions and speeds of both local and remote robots are uniformly bounded.

Since it was proved that $r_m, r_{md}, F_{md}, F_h \in \mathcal{L}_\infty$, it can be seen from (17) that the result $\dot{r}_m \in \mathcal{L}_\infty$ holds. Differentiating (31) once. Then, by noting that $\dot{r}_m, \dot{f}_e^\star \in \mathcal{L}_\infty$ and $|\varepsilon| < 1$ for all $K_e \geq 0$ (by assumption), we have $\dot{r}_s \in \mathcal{L}_\infty$ by using Lemma 1. Since $\dot{r}_s - \dot{r}_{sd} \in \mathcal{L}_\infty$ (see (18)), it is true that $\dot{r}_{sd} \in \mathcal{L}_\infty$. Thus, the control laws $\tau_m$ and $\tau_s$ of the local and remote robots are also uniformly bounded (see (14)–(15)).

Because $\dot{F}_h = -K_h \dot{r}_m + \dot{f}_h^\star$ and $\dot{F}_e = K_e \dot{r}_s + \dot{f}_e^\star$, the result $\dot{F}_h, \dot{F}_e \in \mathcal{L}_\infty$ holds due to the fact that $\dot{r}_m, \dot{r}_s \in \mathcal{L}_\infty$ (as proven above) and $\dot{f}_h^\star, \dot{f}_e^\star \in \mathcal{L}_\infty$ (by assumption). Differentiating (28)–(29) once yields the result $\dot{r}_{md}, \dot{F}_{md} \in \mathcal{L}_\infty$. This therefore completes the proof.

### *3.3. Transparency analysis*

Before conducting the transparency analysis for the closed-loop teleoperation system, we firstly present the following asymptotic convergence result.

**Lemma 4.** *Under the context of Theorem 4, it is true that $\dot{r}_m(t) \to 0$ as $t \to \infty$, i.e., the velocity of the local robot goes to zero asymptotically. Moreover, for all $q \geq 0$, the result $r_m(t) - r_m(t-q) \to 0$ as $t \to \infty$ holds.*

Proof. The most problematic terms in the process of proving the desired asymptotic convergence result are the constant components $d_m$ and $d_s$ of the human force $F_h$ and the environmental force $F_e$, respectively. To get rid of these terms, one can differentiate the closed-loop system (17)–(18) once, which results in

$$\begin{aligned} \ddot{r}_m &= -K_m(\dot{r}_m - \dot{r}_{md}) - \dot{F}_{md} + \dot{F}_h \\ \ddot{r}_s - \ddot{r}_{sd} &= -K_s\left(\dot{r}_s - \dot{r}_{sd}\right). \end{aligned} \tag{38}$$

To analyze the system (38), a new Lyapunov-Krasovskii functional is then proposed as follows

$$V_1 = \frac{1}{2}\dot{r}_m^2 + \frac{1}{2}\left(\dot{r}_s - \dot{r}_{sd}\right)^2 + \frac{1}{2}\int_{t-T}^{t}\left(\dot{u}_{m1}^2 + \dot{u}_{s1}^2 + \dot{u}_{s2}^2\right)d\sigma$$

Note that, $V_1$ is actually the function $V$ shown in (19) when all of the individual functions in $V$ are replaced by their respective derivatives. Differentiating $V_1$ once and using (38) yield

$$\dot{V}_1 = -K_m\left(\dot{r}_m - \dot{r}_{md}\right)^2 - K_s\left(\dot{r}_s - \dot{r}_{sd}\right)^2 + \dot{F}_h\dot{r}_m - \dot{F}_e\dot{r}_s.$$

From (24), it is true that $\dot{F}_h = -K_h\dot{r}_m + \dot{f}_h^\star$ and $\dot{F}_e = K_e\dot{r}_s + \dot{f}_e^\star$, and thus

$$\dot{V}_1 \leq -K_h\dot{r}_m^2 - K_e\dot{r}_s^2 + |\dot{r}_m||\dot{f}_h^\star| + |\dot{r}_s||\dot{f}_e^\star|. \tag{39}$$

By employing the Cauchy inequality, we have

$$\begin{aligned} |\dot{r}_m||\dot{f}_h^\star| &\leq 0.5\alpha_1\dot{r}_m^2 + 0.5\alpha_1^{-1}\dot{f}_h^{\star^2} \\ |\dot{r}_s||\dot{f}_e^\star| &\leq 0.5\alpha_2\dot{r}_s^2 + 0.5\alpha_2^{-1}\dot{f}_e^{\star^2}, \end{aligned} \tag{40}$$

where $\alpha_1, \alpha_2 > 0$ are chosen such that $0 < \alpha_1 < 2K_h$ and that $0 < \alpha_2 < 2K_e$ whenever $K_e > 0$.

In the free-motion mode ($K_e = 0$ and $\dot{f}_e^\star \equiv 0$), from (39)–(40), the result $\dot{V}_1 \leq \left(-K_h + 0.5\alpha_1\right)\dot{r}_m^2 + 0.5\alpha_1^{-1}\dot{f}_h^{\star^2}$ holds. By integrating such a differential inequality once, the following result holds

$$0 \leq -\left(K_h - 0.5\alpha_1\right)\int_0^t \dot{r}_m^2 d\tau + 0.5\alpha_1^{-1}\int_0^t \dot{f}_h^{\star^2} d\tau + V_1(0).$$

Since $V_1(0) \in \mathbb{R}$, $K_h - 0.5\alpha_1 > 0$, and $\dot{f}_h^\star \in \mathcal{L}_2$ (by assumption), the above integral inequality implies that $\dot{r}_m \in \mathcal{L}_2$. Similarly, in the contact mode ($K_e > 0$), substituting (40) into (39) and integrating both sides of the result yield

$$\begin{aligned} 0 \leq &-\left(K_h - 0.5\alpha_1\right)\int_0^t \dot{r}_m^2 d\tau + 0.5\alpha_1^{-1}\int_0^t \dot{f}_h^{\star^2} d\tau \\ &+ 0.5\alpha_2^{-1}\int_0^t \dot{f}_e^{\star^2} d\tau + V_1(0). \end{aligned}$$

Because $\dot{f}_h^\star, \dot{f}_e^\star \in \mathcal{L}_2$ by assumption, it can also be concluded that $\dot{r}_m \in \mathcal{L}_2$ in the contact mode. Thus, in both free-motion and contact scenarios, we always have $\dot{r}_m \in \mathcal{L}_2$. Since $\dot{r}_m \in \mathcal{L}_\infty$ (see the proof of Theorem 4), the result $\dot{r}_m \in \mathcal{L}_2 \cap \mathcal{L}_\infty$ holds. By employing Lemma 3, it is true that $r_m(t) - r_m(t-q) \to 0$ as $t \to \infty$ for all $q \geq 0$.

From Theorem 4, it is known that $\dot{r}_{md}$, $\dot{F}_{md}$, and $\dot{F}_h$ belong to $\mathcal{L}_\infty$. Thus, from (38), the result $\ddot{r}_m \in \mathcal{L}_\infty$ holds. Since $\dot{r}_m \in \mathcal{L}_2$ and $\ddot{r}_m \in \mathcal{L}_\infty$, it is true that $\dot{r}_m(t) \to 0$ as $t \to \infty$, as a result of Barbalat lemma (Farkas and Wegner, 2016). This therefore completes the proof.

The next Theorem demonstrates that the proposed teleoperation system using the force-receptive scattering transformation as shown in Fig. 2 can achieve desired transparency. Recalling that the position tracking error is given by $e_p(t) := r_m(t-T) - r_s(t)$ and also denoting the force tracking error as $e_f(t) := F_h(t) - F_e(t-T)$.

**Theorem 5.** *Consider the closed-loop teleoperation system as described in Theorem 4 and suppose that the assumptions therein remain valid. Then, both position and force tracking errors between the local and remote robots approach zero asymptotically, i.e., $\lim_{t\to\infty} e_p(t) = \lim_{t\to\infty} e_f(t) = 0$.*

Proof. By using Lemma 3 on the assumption that $\dot{f}_e^\star \in \mathcal{L}_2 \cap \mathcal{L}_\infty$, the result $f_e^\star(t) - f_e^\star(t-T) \to 0$

as $t \to \infty$ holds. From Lemma 4, it is true that $\lim_{t\to\infty}[r_m(t) - r_m(t-2T)] = 0$. Therefore, the application of statement 2) of Theorem 3 leads to the fact that $e_p(t) \to 0$, $r_{md}(t) - r_m(t) \to 0$, and $F_{md}(t) - F_e(t-T) \to 0$ as $t \to \infty$. From Lemma 4, it is also known that $\lim_{t\to\infty} \dot{r}_m = 0$, and thus, it can be inferred from (17) that $\lim_{t\to\infty}[F_h(t) - F_{md}(t)] = 0$. Because both $F_{md}(t) - F_e(t-T) \to 0$ and $F_h(t) - F_{md}(t) \to 0$ as $t \to \infty$, the result $\lim_{t\to\infty} e_f(t) = 0$ can immediately be obtained. This therefore completes the proof.

**Remark 3.** Similar to the force-receptive scattering transformation, the generalized scattering transformation introduced in Polushin, 2020 and Usova et al., 2022 also allows transmitting more data independently between the local and remote sides. In comparison with these studies, the proposed force-receptive scattering transformation has the following differences/merits:

- *Teleoperation transparency:* In Polushin, 2020 and Usova et al., 2022, the steady-state transmission requirement was not considered for the formulated scattering transformation scheme, which might result in steady-state errors in both position and force tracking between the local and remote robots. Therefore, the teleoperation transparency was not adequately addressed in these studies.
- *Stability:* The study Usova et al., 2022 did not consider time delay in the formulation where the graph separation stability condition was used. In both Polushin, 2020 and Usova et al., 2022, only finite-gain $\mathcal{L}_2$ stability can be shown for the closed-loop (teleoperation) system. In contrast, the proposed teleoperation system utilizing the force-receptive scattering transformation, under suitable assumptions, can guarantee that: all signals of the closed-loop system are uniformly bounded (Theorem 4), and both position and force tracking errors between the two robots asymptotically converge to zero (Theorem 5). It was also shown that the force-receptive scattering transformation is both BIBO and asymptotically stable (Theorem 3).

In the future, the force-receptive scattering transformation will be extended so that it can actually accept an unlimited amount of input data, similar to what was accomplished in Polushin, 2020 and Usova et al., 2022.

## 4. Comparative simulation result

The proposed teleoperation system using the force-receptive scattering transformation is compared with two notable passivity-based controllers of Nuño et al., 2011 and of Chopra et al., 2008. In Nuño et al., 2011, the control laws of the local and remote robots are

$$\tau_m = -K_{m1}[x_m(t) - x_s(t-T)] - B_m\dot{x}_m$$
$$\tau_s = -K_{s1}[x_s(t) - x_m(t-T)] - B_s\dot{x}_s,$$

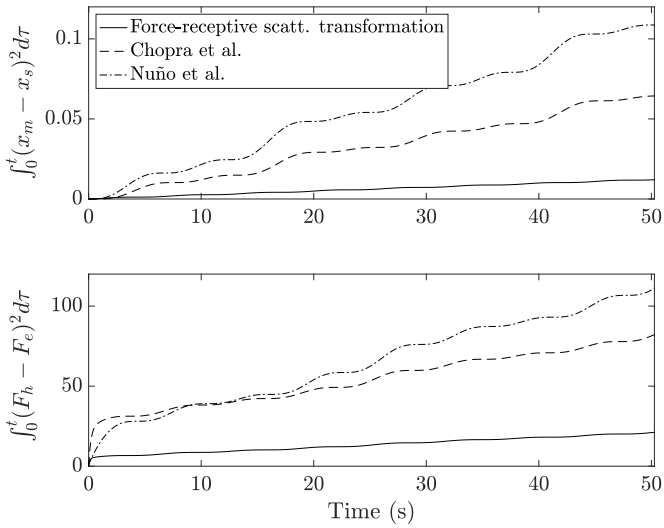


Figure 3: Comparative simulation result.

where $K_{m1}, K_{s1}, B_m, B_s > 0$ and the damping injection must be sufficiently strong, that is to say, the constraint $B_m B_s > T^2 K_{m1} K_{s1}$ needs to be ensured to maintain the system stability. On the other hand, the controllers of Chopra et al., 2008 are given by

$$\tau_m = 0.5b_1[r_s(t-T) - r_m(t)] - \lambda\dot{x}_m$$
$$\tau_s = 0.5b_1[r_m(t-T) - r_s(t)] - \lambda\dot{x}_s,$$

where $b_1 > 0$ is a control gain. Note that these control laws are equivalent to the case when the classical scattering transformation is used, with impedance matching performed on both local and remote sides, and $b_1$ as the characteristic impedance.

The local robot is excited by a non-constant and non-passive force $f_h^\star(t) = 10\sin 0.5t$ N for $0 \le t \le 16\pi$ and $f_h^\star(t) = 0$ N for $t > 16\pi$. Similarly, the remote robot is stimulated by a force $f_e^\star(t) = -5\sin 0.25t$ N for $0 \le t \le 16\pi$ and $f_e^\star = 0$ N for $t > 16\pi$. The remote robot is also assumed to be in contact with an environment that has a moderate stiffness $K_e = 100$. The other force-related parameters are given by $K_h = 20$, $d_m = 0.2$, and $d_s = 0.1$. The time delay is $T = 0.1$ s. The coefficient $\lambda$ in (14)–(15) is chosen as $\lambda = 1$. To draw a fair comparison, all control gains of the three teleoperation architectures are optimized to minimize the following cost function

$$J = \kappa_1 \int_0^{16\pi} (x_m - x_s)^2 \, d\tau + \kappa_2 \int_0^{16\pi} (F_h - F_e)^2 \, d\tau,$$

where $\kappa_1 = 10^3$ and $\kappa_2 = 1$. The particle swarm optimization mechanism is utilized to find the optimal control gains. The optimal results for the architecture of Nuño et al., 2011 are $K_{m1} = 110.44$, $K_{s1} = 113.95$, $B_m = 10.73$, and $B_s = 11.74$, whereas the optimizer for the architecture of Chopra et al., 2008 is $b_1 = 275.51$. In the case of the proposed teleoperation system, the optimal control gains are found as $K_m = K_s = 97.95$, and the optimal scattering weights are $w_{11} = -7.75$, $w_{12} = 0.95$,

$w_{13} = -7.69$, $w_{21} = -6.37$, $w_{22} = -1.90$, $w_{23} = 6.08$, $w_{31} = -0.16$, $w_{32} = -0.15$, $w_{33} = -0.05$, $v_{11} = -6.37$, $v_{12} = -1.90$, $v_{13} = 6.08$, $v_{21} = -6.37$, $v_{22} = -1.90$, $v_{23} = 6.08$, $v_{31} = -0.16$, $v_{32} = -0.15$, $v_{33} = -0.05$.

The simulation result shown in Fig. 3 clearly demonstrates the advantage offered by the force-receptive scattering transformation, i.e., a higher level of teleoperation transparency can be realized if more data is added to the communication protocol between the local and remote sides. Here, the additional piece of information is the actual environmental force, which is independently and intuitively transmitted back to the local side and then employed in the control law of the local robot. This is not possible with the teleoperation schemes of Nuño et al., 2011 and Chopra et al., 2008, which explains the previous finding of the comparative simulation study.

## 5. Conclusions and future work

By increasing the number of data ports and scattering weights accordingly, the force-receptive scattering transformation was designed to handle environmental contact force feedback, which is independently transmitted from the remote side. The newly developed scattering transformation was implemented on a three-channel bilateral teleoperation system, enabling passive data transfer while maintaining the integrity of both transmitted and received signals in the steady state. The effectiveness of the proposed approach in enhancing teleoperation transparency was demonstrated through a comparative simulation study. Our future work is twofold. First, the case of time-varying communication delays will be considered. Second, the proposed force-receptive scattering transformation will be further generalized to accommodate $n$-channel teleoperation systems.


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## Appendix A. Proof of Lemma 2

Since $f$ is assumed to be continuous, in order to demonstrate that $f \in \mathcal{L}_\infty$, we just need to show that $f(t)$ will not escape to infinity as $t \to \infty$. This can be proved by contradiction. Assume that $f(t) \to \infty$ as $t \to \infty$. Let $M > \max\{0, d/K\}$, hence $M > 0$ and $KM - d > 0$. By the definition of infinite limit at infinity (Sherbert and Bartle, 2011, p. 121), there is $\widetilde{T} > 0$ such that for all $t > \widetilde{T}$, we have $f(t) > M$. Hence

$$
\begin{aligned}
&(Kf(\tau) - d)\, f(\tau) > M\,(KM - d) \quad \forall \tau > \widetilde{T} \\
\Longrightarrow &\int_{\widetilde{T}}^{t} (Kf(\tau) - d)\, f(\tau) d\tau > \int_{\widetilde{T}}^{t} M\,(KM - d)\, d\tau \\
\Longrightarrow &\int_{\widetilde{T}}^{t} (Kf(\tau) - d)\, f(\tau) d\tau > M(KM - d)(t - \widetilde{T}).
\end{aligned}
$$

From (34), the following result can be obtained

$$
-\int_0^{\widetilde{T}} (Kf - d)\, f d\tau - \int_{\widetilde{T}}^{t} (Kf - d)\, f d\tau + D \geq 0.
$$

Based on the above inequality, one can be inferred that

$$
M\,(KM - d)\,(t - \widetilde{T}) < -\int_0^{\widetilde{T}} (Kf(\tau) - d)\, f(\tau) d\tau + D.
$$

Therefore, the following result needs to hold for all $t > \widetilde{T}$

$$
t - \widetilde{T} < \frac{1}{M(KM - d)} \left[ -\int_0^{\widetilde{T}} (Kf(\tau) - d)\, f(\tau) d\tau + D \right] \tag{A.1}
$$

Since $f$ is assumed to be continuous, the function $(Kf - d)f$ is Riemann integrable on the interval $[0\ \widetilde{T}]$ (Sherbert and Bartle, 2011, p. 212). For this reason, the integral $\int_0^{\widetilde{T}} (Kf(\tau) - d)\, f(\tau) d\tau$ is finite, and thus, the right side of (A.1) is finite. However, this leads to a contradiction since (A.1) must hold for every $t > \widetilde{T}$. Thus, it was proved that $f(t)$ will not go to infinity as $t \to \infty$. This therefore completes the proof.

## Appendix B. Proof of Lemma 3

We will prove that $\dot{f}(t) - \dot{f}(t-q) \in \mathcal{L}_\infty$ and $f(t) - f(t-q) \in \mathcal{L}_2$, and then employ the Barbalat lemma to conclude the desired result. Firstly, the fact that $\dot{f}(t) - \dot{f}(t-q) \in \mathcal{L}_\infty$ is trivial due to the assumption $\dot{f} \in \mathcal{L}_\infty$. Next, the result $f(t) - f(t-q) \in \mathcal{L}_2$ shall be proven.

Since $f$ is assumed to be differentiable almost everywhere on $\mathbb{R}$ and $f$ is continuous on $\mathbb{R}$, in the sense of generalized Riemann integral, we have

$$
f(t) - f(t-q) = \int_{t-q}^{t} \dot{f}(\tau) d\tau \quad \forall t \geq 0,
$$

which is the result of the fundamental theorem of calculus (Sherbert and Bartle, 2011, p. 295). By applying the Jensen inequality (Fridman, 2014, p. 87), the following result holds

$$
\begin{aligned}
&[f(t) - f(t-q)]^2 = \left( \int_{t-q}^{t} \dot{f}(\tau) d\tau \right)^2 \leq q \int_{t-q}^{t} \dot{f}^2(\tau) d\tau \\
\Longrightarrow &\int_0^t [f(\tau) - f(\tau - q)]^2\, d\tau \leq q \int_0^t \int_{\tau-q}^{\tau} \dot{f}^2(\sigma) d\sigma d\tau.
\end{aligned} \tag{B.1}
$$

The right side of (B.1) can be evaluated by denoting

$$
R = \int_0^t \int_{\tau-q}^{\tau} \dot{f}^2(\sigma) d\sigma d\tau.
$$

Such an integral exists since $\dot{f}$ is supposed to be uniformly bounded and continuous almost everywhere on $\mathbb{R}$

(Lebesgue integrability condition (Sherbert and Bartle, 2011, p. 221)). By changing the order of integration, the repeated integral $R$ can be decomposed into a sum of single integrals as follows

$$R = R_1 + R_2 + R_3,$$

where $R_i$ $(i \in \mathcal{N})$ are given by

$$R_1 = q\int_0^t \dot{f}^2(\sigma)d\sigma,\ R_2 = \int_{-q}^0 (\sigma+q)\dot{f}^2(\sigma)d\sigma$$
$$R_3 = \int_{t-q}^t (t-\sigma-q)\dot{f}^2(\sigma)d\sigma.$$

The upper bounds of $R_i$ $(i \in \mathcal{N})$ will be estimated next. By assumption, $\dot{f} \in \mathcal{L}_2$, and thus there is $M_1 > 0$ such that $R_1 < M_1\ \forall t \geq 0$. It is also supposed that $\dot{f} \in \mathcal{L}_\infty$, hence there exists $M_2 > 0$ so that $|\dot{f}(t)| < M_2\ \forall t \geq 0$. Consequently

$$R_2 = \int_{-q}^0 (\sigma+q)\dot{f}^2(\sigma)d\sigma = \int_0^q \tau\dot{f}^2(\tau-q)d\tau$$
$$< \int_0^q |\tau||\dot{f}^2(\tau-q)|d\tau < M_2^2\int_0^q |\tau|d\tau = 0.5\,(M_2 q)^2\,.$$

We also have that

$$R_3 = \int_{t-q}^t (t-\sigma-q)\dot{f}^2(\sigma)d\sigma = -\int_0^q \tau\dot{f}^2(\tau+t-q)d\tau$$
$$< \int_0^q |\tau||\dot{f}^2(\tau+t-q)|d\tau = M_2^2\int_0^q |\tau|d\tau = 0.5(M_2 q)^2.$$

In summary, it is true that

$$R < M_1 + (M_2 q)^2 < \infty \quad \forall t \geq 0.$$

Hence, according to (B.1), we have $f(t) - f(t-q) \in \mathcal{L}_2$. Combining this result with the fact that $\dot{f}(t) - \dot{f}(t-q) \in \mathcal{L}_\infty$, it can be inferred that $f(t) - f(t-q) \to 0$ as $t \to \infty$ by Barbalat lemma (Farkas and Wegner, 2016). This therefore completes the proof.

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